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(12) **United States Plant Patent**
Leis

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(54) **STRAWBERRY PLANT NAMED ‘CIVH413’**

(50) Latin Name: *Fragaria x ananassa* Duchesne ex
Rozier
Varietal Denomination: **CIVH413**

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See application file for complete search history.

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(57) **ABSTRACT**

This invention relates to a new and distinct variety of strawberry plant named ‘CIVH413’. The new June bearing strawberry variety named ‘CIVH413’ is primarily adapted to the climate and growing conditions of continental climate and is characterized by a rustic plant with upright habit and visible fruit easy to collect; the vigour is medium. The new strawberry has a high productivity, consistent during all the picking season; in particular is tolerant to the most common leaf and root diseases. The fruits have conical, attractive and very regular shape with orange-red colour; the size is big and consistent during all the time of picking. The firmness is high with a very good shelf life. The flavour is very good, balanced and aromatic.

4 Drawing Sheets

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Latin name of the genus and species of the plant claimed:
Fragaria x ananassa Duchesne ex Rozier.
Variety denomination: ‘CIVH413’.

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority from Community Plant
Variety Rights Application No. 2021/2768, filed Nov. 4,
2021.

BACKGROUND OF THE INVENTION

The present invention relates to a new and distinct straw-
berry variety named ‘CIVH413’. The variety is botanically
known as *Fragaria x ananassa* Duchesne ex Rozier.

The new strawberry ‘CIVH413’ is a product of a planned
breeding program conducted by the inventor, Michelangelo
Leis in San Giuseppe di Comacchio, Ferrara, Italy. The
objective of the breeding program is to develop june bearing
strawberry variety with high chill requirement, high and
constant productivity during all the picking season; fruit
with very regular and conical shape, easy to collect, big and
constant fruit size, high firmness and very good shelf life.
Rustic plant, tolerant to the most common leaf and root
diseases.

This new strawberry ‘CIVH413’ is a result of a controlled
cross made by the inventor in 2011, in San Giuseppe di
Comacchio, Ferrara, Italy. The male or pollen parent is
strawberry selection designated 6C9C-007 (unpatented pro-

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prietary selection). The female or seed parent is strawberry
selection designated L7H3-017 (unpatented proprietary
selection).

The new strawberry ‘CIVH413’ was discovered and
selected by the inventor as a single flowering plant within
the progeny of the stated cross in May 2013 in San Giuseppe
di Comacchio, Ferrara, Italy (44°45' North and 12°11' East).
After its selection, the new variety was asexually propagated
by stolons in a nursery located in San Giuseppe di Comac-
chio, Ferrara, Italy. The new variety was extensively tested
over the next several years in different European area with
high chill condition. This propagation has demonstrated that
the combination of characteristics as herein disclosed for the
new cultivar are firmly fixed and retained through successive
generations of asexual reproduction. The new cultivar repro-
duces true to type.

BRIEF SUMMARY OF THE INVENTION

‘CIVH413’ is primarily adapted to the climate and grow-
ing with high chill condition. This area provides the needed
and correct temperatures for flower induction, and to pro-
duce a strong and vigorous plant, and to maintain fruit
quality during full production.

The following traits have been repeatedly observed and
are determined to be unique characteristics of ‘CIVH413’,
which in combination distinguish this strawberry plant as a
new and distinct variety:

1. Plants with upright habit, with clearly visible fruit.
2. Fruits are easy to collect.

3. Fruit with conical, attractive and very regular shape.
4. 'CIVH413' has high firmness and very good shelf life.
5. Post-harvest fruit longevity—6-8 days, if stored according to industry standards.
6. High productivity, consistent during all the picking season.
7. Tolerant to most common leaf and root diseases.

Plants of the new strawberry variety 'CIVH413' differ from plants of the parents, 6C9C-007 and L7H3-017 in the characteristics described in Table 1.

TABLE 1

Characteristic	'CIVH413'	6C9C-007	L7H3-017
Time of ripening	Early	Medium	Early
Growth habit	Upright	Upright	Semi-upright
Plant vigour	Medium	Medium to strong	Strong
Fruit size	Large	Medium	Large to very large
Firmness	High	Medium to high	Medium

Of the many commercial cultivars known to the present inventor, the most similar in comparison to the new strawberry variety 'CIVH413' is the strawberry variety 'CLERY' (unpatented). Plants of the new strawberry variety 'CIVH413' differ from plants of strawberry variety 'CLERY' (unpatented) in the characteristics described in Table 2:

TABLE 2

Characteristic	'CIVH413'	'CLERY' (unpatented)
Fruit: time of beginning ripening	Early	Very early
Fruit: colour	Orange red	Medium red
Plant: growth habit	Upright	Semi-upright

BRIEF DESCRIPTIONS OF THE PHOTOGRAPHS

The accompanying color photographs illustrate the overall appearance of typical specimens of the new strawberry variety 'CIVH413', at various stages of development as true as it is reasonably possible with color reproductions of this type. Color in the photographs may differ slightly from the color value cited in the botanical description which accurately describe the color of 'CIVH413'. The depicted plant and plant parts of the new strawberry variety 'CIVH413' were taken in San Giuseppe di Comacchio, Ferrara, Italy, and are approximately 6 to 9 months old.

FIG. 1 shows plants of 'CIVH413';

FIG. 2 shows the leaves of 'CIVH413';

FIG. 3 shows typical flowers of 'CIVH413'; and

FIG. 4 shows typical fruits of 'CIVH413'.

DETAILED BOTANICAL DESCRIPTION

'CIVH413' has not been observed under all possible environmental conditions. The characteristics of the new variety may vary in detail, depending upon variations in environmental factors, including weather (temperature, humidity and light intensity), day length, soil type and location.

The aforementioned photographs, together with the following observations, measurements and values describe the new strawberry variety 'CIVH413', unless otherwise noted, taken during 2022 growing season in San Giuseppe di Comacchio, Ferrara, Italy. The observations, measurements and values were taken from plants of 'CIVH413' dug in December 2020 from a nursery located in 44021 Codigoro, Ferrara, (locality Pomposa), Italy, and planted in July 2021 in a test field in San Giuseppe di Comacchio, Ferrara, Italy.

From December 2020 to July 2021, the plants were stored in cold room, in order to undergo the hours of cold necessary for the correct development of the plants. Plants of the new strawberry variety 'CIVH413' were grown under conditions which closely approximate those generally used in commercial practice.

The observed plants were one year old plants. The plants used in the production field are produced in a nursery located in 44021 Codigoro, Ferrara, (locality Pomposa), Ferrara, Italy.

Yield observations and fruit quality characteristics are averaged from 3 years of data collected from the 2020 through 2022 growing seasons. Flower measurements and characteristics are from secondary flowers unless otherwise noted. Fruit characteristics and measurements are from secondary fruit unless otherwise noted.

Color references are made to The 1986 Royal Horticultural Society Colour Chart (R.H.S.), except where general colors of ordinary significance are used. Color values were taken under daylight conditions approximately in the afternoon in San Giuseppe di Comacchio, Ferrara, Italy. The age of the observed plants is one year old.

The following tables 3-9 describe fruit, plant, stolon, foliage, fruiting truss, flower and pest/disease characteristics of the new strawberry 'CIVH413'.

TABLE 3

FRUIT CHARACTERISTICS	
Characteristic	'CIVH413'
Color of mature fruit	Red group - RHS 44B
Color of internal flesh	Red group - RHS 41B
Color of internal core	Red group - RHS 41B
Length (cm)	About 5.0 cm
Width (cm)	About 3.5 cm
Ratio length/width	Moderately longer
Calyx diameter (cm)	About 3.4 cm
Average weight (gm)	About 26.10 g
Achene color	Yellow-green group 150C
Number of achenes per cm ² measured at the center of berry	About 7
Average weight of 1000 achenes (g)	About 0.80 g
Marketable yield (gm/plt)	Average 1050 gm/plt
Size	Large
Predominant shape	Conical
Difference in shapes between primary and secondary fruit	Slight
Band without achenes	Narrow
Unevenness of surface	Even or slightly uneven
Evenness of color	Strong
Glossiness	Strong
Insertion of achenes	Level with surface
Insertion of calyx	Level with fruit
Attitude of the calyx	Outward
Size of calyx in relation to fruit diameter	Smaller
Adherence of calyx	Strong
Firmness of skin	Very firm

TABLE 3-continued

FRUIT CHARACTERISTICS	
Characteristic	‘CIVH413’
Firmness of flesh	Very firm
Distribution of red color of the flesh	Uniform
Hollow center expression	Absent or small
Flavor	Very good flavour, balanced and aromatic
Soluble solids (% brix)	About 7.4%
Time of first flowering	Early. The full bloom of first flowering is at the end of march, in North Italy - Po Valley
Time of first harvesting	Early. The first harvest period is from the beginning of May to the end of May, in North Italy - Po Valley
Harvest period	From the middle of May to the beginning of June in San Giuseppe di Comacchio, Ferrara, Italy
Type of bearing	Not remontant

TABLE 4

PLANT CHARACTERISTICS	
Characteristic	‘CIVH413’
Height (cm)	About 20 to 25 cm
Spread (cm)	About 30 to 40 cm
Size	Medium
Habit	Upright
Density	Medium
Vigor	Medium

TABLE 5

STOLON CHARACTERISTICS	
Characteristic	‘CIVH413’
Average number per plant	About 10-15
Typical and Average length	40 (cm)
Anthocyanin coloration	Red group - 53B
Anthocyanin intensity	Medium
Diameter at bract (mm)	About 3 mm
Pubescence	Sparse to medium

TABLE 6

FOLIAGE CHARACTERISTICS	
Characteristic	‘CIVH413’
Foliage:	
Color of upper surface	Green group - 137A
Color of underside	Green group - 138B
Shape in cross section	Concave
Interveinal blistering	Medium
Glossiness	Medium
Number of leaflets	3
Entire leaf size:	
Length:	About 12.5 cm
Width	About 16 cm

TABLE 6-continued

FOLIAGE CHARACTERISTICS	
Characteristic	‘CIVH413’
Terminal Leaflet:	
Length (cm)	About 9.0 cm
Width (cm)	About 8.1 cm
Length/width ratio	Equal
Serrations/leaf	Overlapping
Size	Medium
Shape of base	Obtuse
Shape of teeth	Acute to rounded
Petiole:	
Color of petiole	Yellow-Green group - 144B
Length (cm)	About 24.5 cm
Diameter (mm)	About 0.4 cm
Pubescence	Medium
Attitude of hairs	Slightly outwards
Stipule:	
Shape of stipule	narrow-elliptic
Length (mm)	About 2.85 cm
Width (mm)	About 9 mm
Anthocyanin coloration	Absent or very weak (Greyed-Purple-186B)
Color	Yellow-green group 145C

TABLE 7

FRUITING TRUSS CHARACTERISTICS	
Characteristic	‘CIVH413’
Length (cm)	About 20-25 cm
Attitude	Erect
Position relative to foliage	Same level
Number of fruit per truss	1 to 5
Pubescence	Strong
Anthocyanin intensity	Absent or very weak (Greyed-Purple-186B)
Attitude at first pick	Horizontal

TABLE 8

FLOWER CHARACTERISTICS	
Characteristic	‘CIVH413’
Petal color	
Mature (upper)	White group - 155D
Mature (lower)	White group - 155D
Petal shape	
Overall	Circular
Apex	Rounded
Base	Rounded
Petal length (mm)	About 1 to 1.2 cm
Petal width (mm)	About 1 to 1.2 cm
Petal length/width ratio	Equal
Number of petals/flower	About 5 to 6
Sepals color	
Mature (upper)	Green group - 137A
Mature (lower)	Green group - 138A
Sepal shape	
Overall	Elliptic
Apex	Pointed
Base	Flat
Sepal length (mm)	About 0.7 to 1.2 cm
Sepal width (mm)	About 0.3 to 0.5 cm

TABLE 8-continued

FLOWER CHARACTERISTICS	
Characteristic	‘CIVH413’
Sepal length/width ratio	Longer
Number of sepals/flower	About 9 to 13
Corolla diameter (mm)	About 21 mm
Calyx diameter (mm)	About 28 mm
Size of calyx relative to corolla	Larger
Size of inner calyx relative to outer calyx	Smaller
Relative position of petals	Touching

TABLE 9

REPRODUCTIVE ORGANS	
Characteristic	‘CIVH413’
Receptacle color	Yellow green group - 154B
Pollen color	Yellow orange group - 14A
Stamen	Present
Stamen:	Numerous
Length:	Approximately 2.7 mm. Color: RHS 144 D yellow-green group
Anthers:	Generally average in size (length: about 1.0 mm, width: about 0.75 mm). Color: RHS 15 B yellow orange group and darkening with advanced maturity.

TABLE 9-continued

REPRODUCTIVE ORGANS	
Characteristic	‘CIVH413’
Pistils:	Numerous, generally average in size. Color: 151 D green group.
Pollen amount:	Much
Fruit use:	Fresh consumption
Shipping quality:	High

TABLE 10

PEST AND DISEASE REACTIONS	
Characteristic	‘CIVH413’
Powdery mildew (<i>Sphaerotheca macularis</i>)	Moderately tollerant
Angular leaf spot (<i>Xanthomonas fragariae</i>)	Moderately susceptible
Botrytis fruit rot (<i>Botrytis cinerea</i>)	Very low sensibility
Fusarium wilt (<i>Fusarium oxysporum</i>)	Moderately susceptible
Anthracnose crown rot (<i>Colletotrichum fragariae</i>)	Moderately susceptible
Two-spotted spider mite (<i>Tetranychus urticae</i>)	Susceptible

I claim:
1. A new and distinct strawberry plant named ‘CIVH413’,
as herein described and illustrated.

* * * * *

FIG. 1

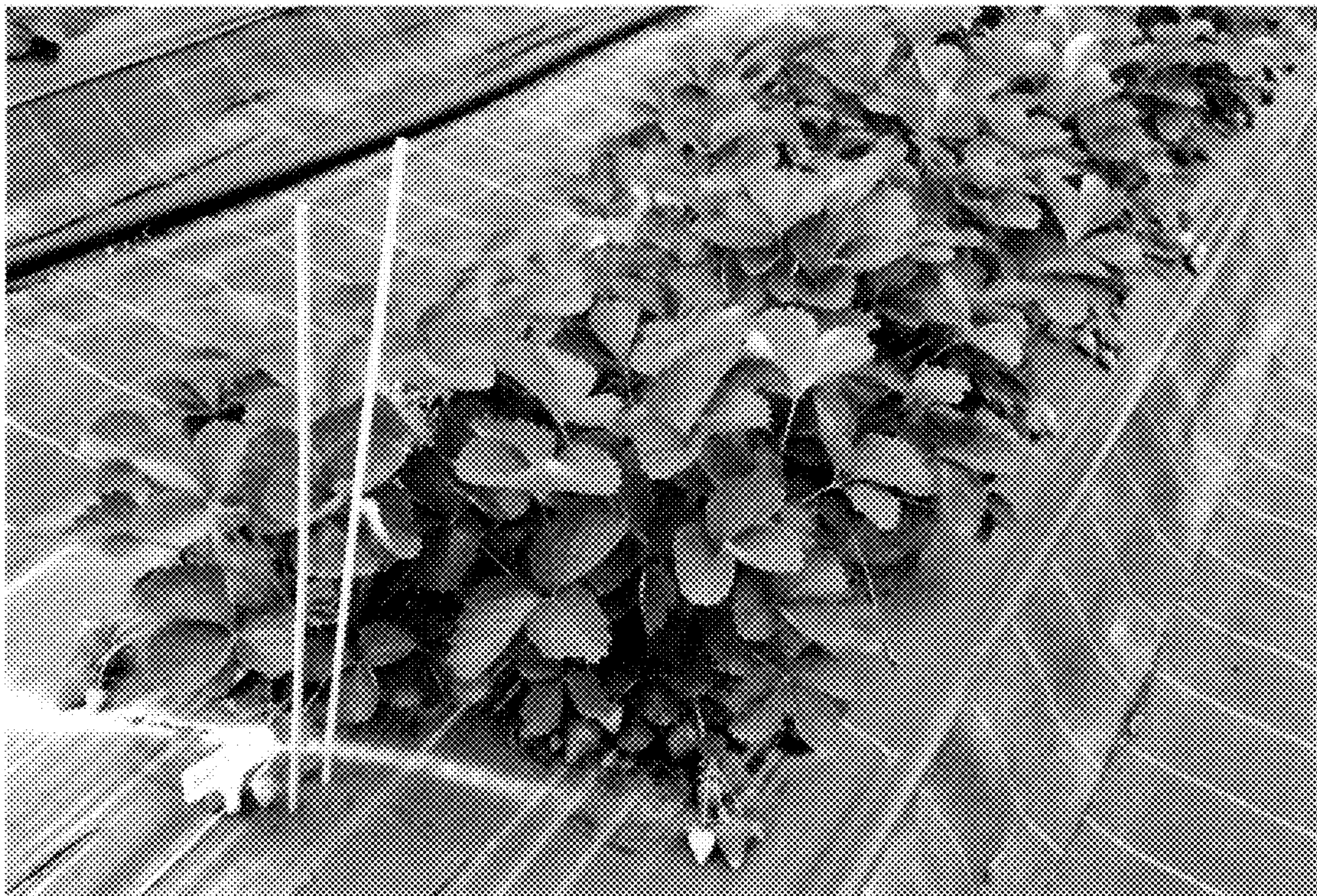


FIG. 2

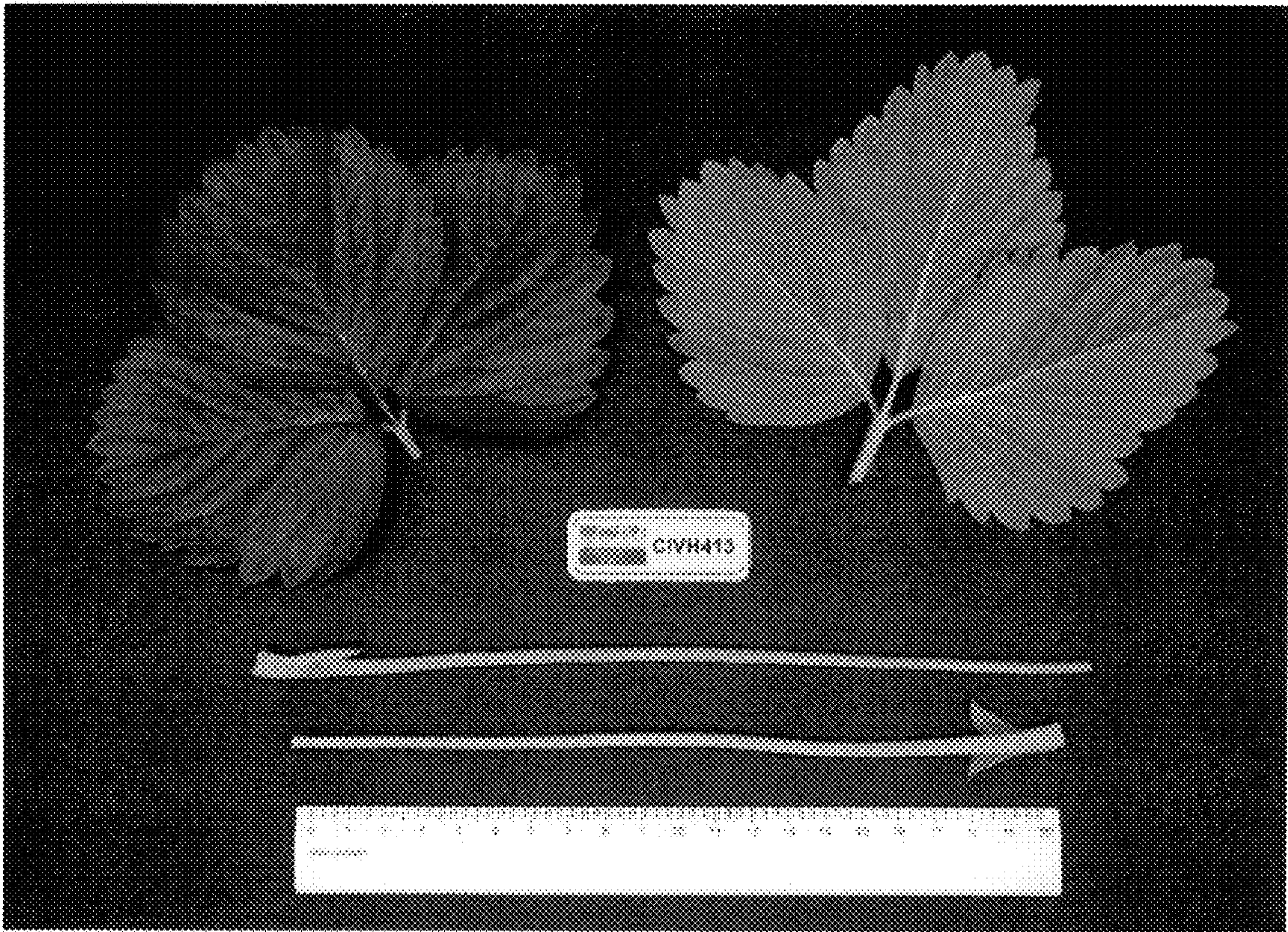


FIG. 3

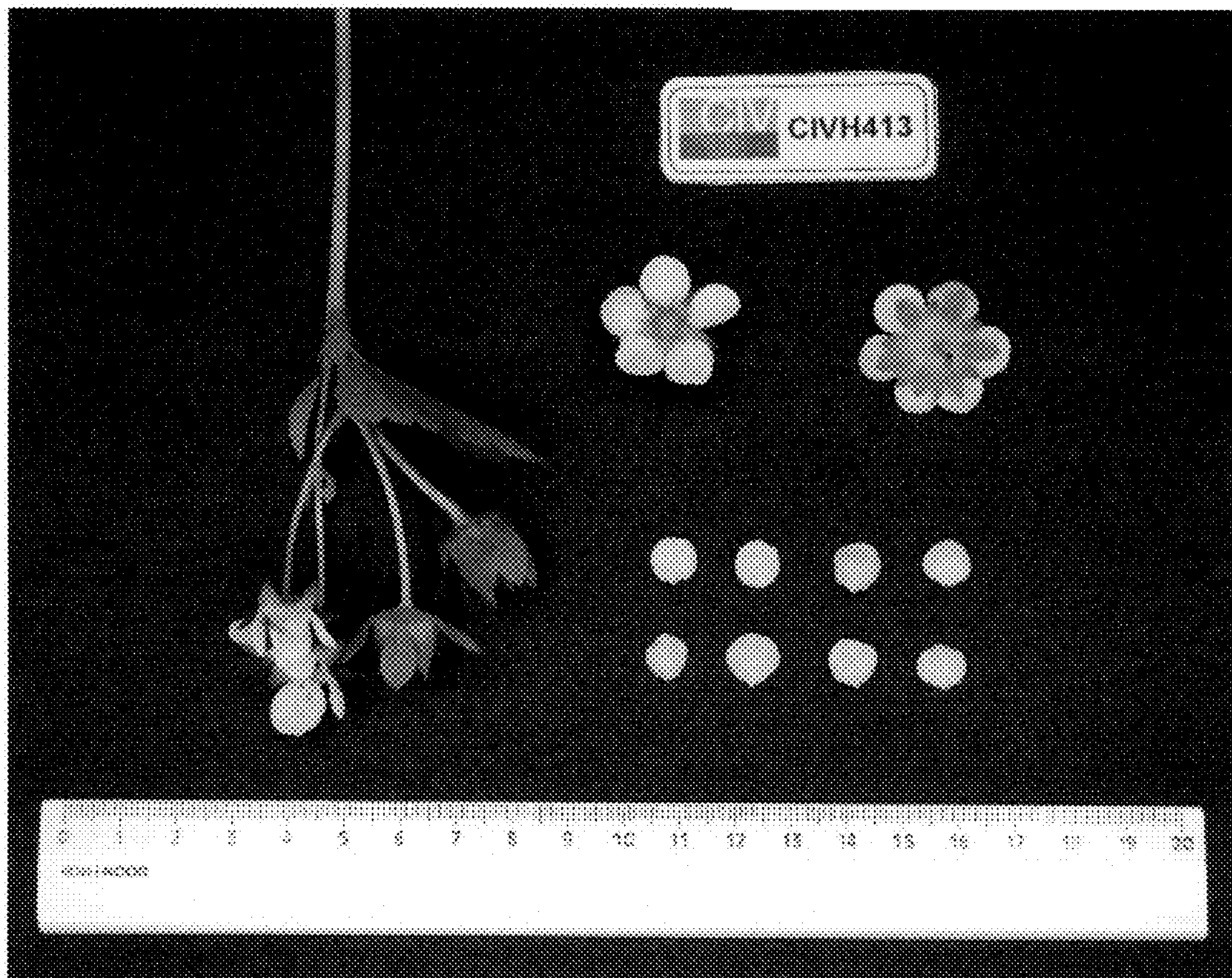


FIG. 4

